

DINESH R

Software Development Engineer (Backend + AI Systems)

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EXPERIENCE

Software Development Engineer Intern

Jun 2026 – Present

MiddleBank Consulting Group

Auckland, New Zealand (Remote)

- Architecting an **end-to-end AI warehouse workflow orchestration system** using **LangChain, FastAPI, and Next.js** to automate inventory tracking, purchase orders, and reporting across finance, supply chain, and retail verticals improving efficiency by **42%+**
- Developing and deploying **AI-powered full-stack applications** for cross-domain business operations, integrating **LLM-driven decision engines** with modern web interfaces to streamline workflows and accelerate decision-making across multiple group entities

Software Development Engineer Intern

Dec 2025 – May 2026

TripFactory.com

Bengaluru, India (On-site)

- Engineered an **AI-powered workflow automation** using **Spring Boot and LangChain4j** to process **200+ tour packages** from Excel into the booking platform, cutting data entry time from **3 days to 10 minutes** and eliminating **95% of manual effort**
- Developed and maintained **Core Java (Spring Boot) backend services** supporting high-volume internal workflows for travel package management, ensuring system reliability and data consistency across booking pipelines

Software Development Engineer Intern

Jun 2025 – Nov 2025

Thoothukudi Police – Cybercrime Branch

Thoothukudi, India (Hybrid)

- Built and live-deployed a **predictive crowd density system** using **XGBoost and Scikit-learn** on CCTV feeds and historical data across **5+ zones**, cutting deployment costs by **40%** and congestion by **60%**
- Engineered a **smart parking allocation platform** using **Next.js and machine learning pipelines** with real-time slot availability and vehicle arrival prediction, reducing manual coordination overhead by **45%** across live event deployments

PROJECTS

GlobalXport – AI-Powered International Trade Automation Platform

- Automated end-to-end generation of **5+ export documents**, compressing processing time from **7 days to 4 hours** using multi-agent LLM pipelines
- Designed **modular microservices architecture** with independent document agents; integrated **multi-LLM orchestration** (Gemini, ChatGPT, Claude) with Pinecone vector search for compliance validation, improving efficiency by **35%**
- **Tech Stack:** Next.js, PyTorch, Python, FastAPI, MongoDB, Pinecone, GCP, LangChain, Multi-LLM

AlloyMind AI – Intelligent Metallurgical Optimization Platform

- Built **ensemble ML system** (LightGBM + CatBoost) achieving **92.5% prediction accuracy** for alloy composition optimization across EN1563, ASTM A536, and ASTM A395 industrial grades
- Engineered **FastAPI + Next.js** full-stack platform with real-time analytics, cost-impact analysis, and RBAC; implemented priority-ranked recommendation engine with **sub-100ms** response time
- **Tech Stack:** Next.js, TypeScript, FastAPI, Python, MongoDB, LightGBM, CatBoost, Scikit-learn

EDUCATION

Bannari Amman Institute of Technology

Sep 2023 – Apr 2027

B.Tech Information Technology | CGPA: 8.8 / 10.0

Tamil Nadu, India

- **Coursework:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Generative AI, System Design

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript, SQL

Backend & AI: Spring Boot, FastAPI, Flask, LangChain, PyTorch, RAG, LLM Orchestration, Prompt Engineering, LLM Ops

Frontend: React.js, Next.js, Tailwind CSS

Cloud & DB: AWS, MySQL, PostgreSQL, MongoDB, Supabase, Redis, FAISS, Pinecone, Vector Databases

DevOps: Git, Docker, Linux, Postman, CI/CD, Unit Testing, Microservices, REST APIs

ACHIEVEMENTS & RECOGNITION

- **Zafin Aspire & Achieve Global Merit Scholarship (2025–26)** – **Top 4 engineering student globally** from 5,000+ applicants; awarded **\$5,000 USD** + global industry internship
- **2nd Runner-Up, ABB EngineeredX 2025** – Best **AI agentic automation** for manufacturing among **500+ teams** across South India
- **2nd Runner-Up, Blueprint @ IIT Delhi 2025** – Only student prototype in Top 3 among **100+ startups**
- **Top 50 Campus Startups** – Google for Startups × Campus Fund 2025; awarded **\$250,000 Cloud Credits**
- **Leadership:** AI Innovation Cell (Tech Executive Lead) | Rotaract Club (Innovation Lead) | Public Speaker

CERTIFICATIONS

- **Google AI & Prompting Essentials** – AI workflows, prompt engineering, chain of thought, LLM optimization ([Google](#))
- **GitHub Foundations** – Version control, CI/CD, collaborative workflows ([GitHub](#))